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Final Report

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1. General Information

Project Name: BlitzCook

Value Proposition: Learn from your mistakes!

Group Name: STRFLS

Group Members:
- De Dominicis Francesco Pio
- Pantani Pietro
- Tornesello Gianluca

2. Problem/Solution Overview

Beginner Amateur Cooks can be overwhelmed by new recipes and have a hard time learning certain steps of said recipes. The only way to learn these specific steps is with a lot of trial and error, resulting in waste of ingredients.

Our solution is based around the idea of **Shock Therapy**: the user is put in a critical situation and needs to manage it in the best possible way. The application generates new exercises thanks to AI, starting from a prompt describing what problem the user wants to learn to avoid, enabling him to always have the right exercise for their needs.

3. Needfinding

3.a Domain of Interest

The domain of interest we've selected is "Beginner Amateur Cooks". This was the outcome of a significant brainstorming session. Among all the possible themes considered, this was the one that highlighted the highest potential to develop into something useful for everyday life.

This specific target was chosen also because we noticed that most of the issues related to cooking were relatable to inexperience. Therefore, we aimed to create something meaningful for those who approach this world for the first time.

3.b Interviews

3.b.i PARTICIPANTS

INTERVIEW 1

Immediate User
Samuele Pagano, 25 yo
Primary School Teacher
Onsite

Interviewers
- *Gianluca Tornesello (leading)*
- *Pietro Pantani (note taker)*

INTERVIEW 2

Immediate User
Andrea Contardi, 22 yo
Student
Online

Interviewers
- *Gianluca Tornesello (leading)*
- *Pietro Pantani (note taker)*

INTERVIEW 3

Immediate User
Matteo Cappellari, 20 yo
Polito Student
Onsite

Interviewers
- *Francesco Pio De Dominicis (leading)*
- *Pietro Pantani (note taker)*

INTERVIEW 4

Extreme User
Giovanni Cranco, 19 yo
Culinary Education Student
Online

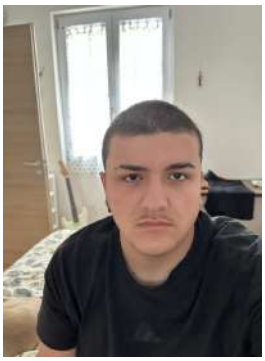
Interviewers
- *Pietro Pantani (leading)*
- *Francesco Pio De Dominicis (note taker)*
- *Gianluca Tornesello (note taker)*

INTERVIEW 5

Extreme User
Giovanni Cranco, 17 yo
Culinary Education Student
Online

Interviewers
- *Pietro Pantani (leading)*
- *Francesco Pio De Dominicis (note taker)*
- *Gianluca Tornesello (note taker)*

INTERVIEW 6



Extreme User
Diego Garzia, 19 yo
Culinary Education Student
Online

Interviewers
- *Gianluca Tornesello (leading)*
- *Francesco Pio De Dominicis (note taker)*
- *Pietro Pantani (note taker)*

INTERVIEW 7

Domain Expert
Adriana Manzo, 51 yo
Food Blogger
Online

Interviewers
- *Pietro Pantani (leading)*
- *Gianluca Tornesello (note taker)*
- *Francesco Pio De Dominicis (note taker)*

3.b.ii METHODOLOGY & PROCEDURE

We conducted a total of seven interviews, all in Italian, both in-person and remotely depending on the availability of our interviewees. Recruitment was based on direct and indirect contacts with the aim of finding beginner amateur cooks to understand their approach to cooking and their habits.

We interviewed three students as **Immediate Users**. Regarding **Extreme Users**, our research focused on culinary education students who, unlike the immediate users, study daily to achieve the goal of working in the sector. Therefore, they were able to offer a different perspective, more oriented toward the profession. Eventually, for the **Domain Expert** we interviewed the owner of a cooking blog with years of experience in the sector, who interacts daily with amateur cooks using her blog to learn new recipes and improve.

All participants signed a **consent form**, regarding both the recording and the subsequent processing of information gathered during the interview. We prepared two different versions, one for underage interviewees, requiring the signature of a parent, and one for adults. Both versions were written both in English and in Italian.
The consent forms can be accessed at the following link:
https://github.com/polito-hci-2025/STRFL/tree/main/A1/ConsentForms

We started with standard questions to allow participants to offer topics for further exploration through follow-up questions, except for the domain expert, where we wanted to delve into her specific vision, how cooks interact with her blog, and the perspective of a more mature individual.

Team members switched roles through interviews, with each being interviewer or note taker at least once.

Materials used were the same for all the interviews: smartphone, computer, audio recording program, notebook, pen.

QUESTIONS

After a brief introduction and explanation of the interview, and after asking for consent to record, we asked standard general questions to all users. In contrast, for the domain expert, we opted for questions regarding her blog and her perspective on the users who visit it.

USER QUESTIONS

INTRODUCTORY QUESTIONS

- How did you first get interested in cooking?
- How long have you been cooking?
- What is your level of expertise?
- Do you remember a moment or a period when you started cooking more often?

CHOOSING THE RECIPE

- Can you tell me about the last time you looked for a new recipe?
- Tell me about how you usually decide what to cook. What influences your choice?
- Tell me about a time when you had decided to cook something but later changed your mind. What made you change your decision?

PREPARING THE DISH

- Tell me about the last thing you cooked: how do you organize the preparation?
- Thinking about your cooking routine, what tools do you use – both digital and non-digital?
- Tell me about a time when a recipe didn't turn out as you expected. What went wrong?
- What is your approach to food waste?
- If you could change something about your daily cooking experience, what would you change first?

CONCLUSION

- Is there any aspect of the recipe-making process that we haven't talked about, but that you'd like to mention?

DOMAIN EXPERT QUESTIONS

When these recipes are chosen, recorded, and uploaded, how important is the difficulty of the recipe in the selection process? Do users often complain about difficulties in preparation?

- Have you ever had to adapt recipes for less experienced cooks, perhaps thinking there were valid alternatives for other reasons, such as ingredients?

- What tools are used for preparations, both digital and non-digital, if any?
- In your opinion, what are the differences between a amateur cook and a more experienced one? Are there trends or differences even in the preference for preparations or ingredients?
- For amateurs, what are the biggest challenges to face? Where do the greatest difficulties lie?
- How is food waste generally managed in the kitchen? Are there methods you try to introduce to manage this waste?
- How are these recipes generally received by most people?
- Has it ever happened that during the preparation of a recipe something went wrong, so it turned out badly, and if so, what happened?
- In your opinion, was there also a way to potentially prevent this in this example (of the forgotten yeast), perhaps?
- In your opinion, is there anything I haven't asked you that could be a tip for amateurs to improve over time?

FOLLOW UP QUESTIONS

Moreover, we asked some follow-up questions to each volunteer according to the insights he/she provided us.

Andrea Contardi

- *"And before we move on from the topic of following recipes, do you ever happen to invent a recipe, essentially just having fun in the kitchen?"*
- *"In your opinion, what do you generally do to try and improve? You mentioned your current level; what steps do you take to try and reach the next level?"*
- *"So, do you put particular emphasis on initial organization to make things easier for yourself?"*
- *"In your opinion, how could you prevent the recipe from turning out poorly in this instance?"*
- *"And why do you believe that social media videos might be more useful than just reading the instructions?"*

Diego Garzia

- *"You mentioned that you attended culinary school (Istituto Alberghiero). I wanted to ask, how did that experience help and prepare you for the culinary world? Do you feel the school provided you with all the tools you need?"*
- *"Earlier you told me that culinary school didn't give you too much theoretical knowledge, or rather, certainly not strictly practical knowledge. How did you make-up for these gaps?"*
- *"How could you have prevented this mistake?"*

Francesco Cranco

- *"You already mentioned you attend culinary school; apart from that, have you ever had any other type of training?"*
- *"And regarding changing ingredients or techniques—has this ever happened while you were already working, meaning once the recipe had already started?"*
- *"In your opinion, what is the most complicated part of this entire process?"*
- *"Now that you've told me about scraps, how do you generally manage them, aside from the ones that obviously have to be thrown away?"*
- *"Sorry, I find that interesting. How specifically does this course help you? In what way precisely? And is there anything it doesn't help you with?"*

Giovanni Cranco

- *"Are there other characteristics a recipe must have for you to see it and say, 'Okay, I want to make this'?"*
- *"You spoke about very advanced techniques... how do you generally learn them?"*
- *"Speaking of which, do you often cook for other people as well?"*
- *"...Do these advanced techniques ever feel like a hurdle when you read a recipe or think of something, or do you tackle them as if they were nothing?"*

Matteo Cappellari

- *"And for how long, actually? Do you consider yourself an amateur cook?"*
- *"Do you feel you've had problems in the past using these tools?"*

Samuele Pagano

- *"And what difficulties do you encounter when approaching a new recipe?"*
- *"And does it ever happen that a recipe requires a tool you don't have at home? What do you do in that case?"*
- *"Generally, how many people do you cook for? If you happen to cook for more people, for example, how do you manage the quantities?"*
- *"When choosing what to cook daily, what do you base your decision on? Do you follow a certain diet, or is it based on what you have at home?"*
- *"You said you often cook for friends. How do you handle the different dietary necessities of many people?"*

3.b.iii RESULTS

During the interviews, we took on-the-fly notes about the insights that we believed relevant.

Here are the key quotes that we found in a first instance.

Interview 1 - Samuele Pagano

"I often look for them [recipes] on TikTok or YouTube, ***because watching the preparation process helps me a lot, even more than reading****."*

"I remember that initially ***the main difficulties were timing and cleaning****."*

"So, ***regarding the time problem, I think the solution is experience: cooking a lot and really understanding the timing and the necessary steps****..."*

"It has happened that I've burned something I forgot in the oven. ***In that case, unfortunately, if something is burnt, it's really burnt; there is nothing left to do*** *but throw the dish away."*

"I ***try not to waste food through prevention****, like checking all the labels beforehand. Ever since that happened, I always do it."*

"I am very careful about waste because I hate wasting food and money, obviously. ***So, before starting a recipe where I have to go buy ingredients, I try to study it as best as I can through various sources****."*

"If I'm missing an ingredient, ***I search on the internet, even using Artificial Intelligence like ChatGPT, or other cooking sites.*** *I search, for example: 'How can I substitute the two eggs in this recipe?'"*

*"****One thing that might change my mind is the cost of preparation****. If I realize that a certain dish costs a lot... maybe I went to the supermarket intending to make it, but I realize I'm spending a ton of money, I say 'No, never mind, let's forget it,' and I change my mind."*

Interview 2 - Andrea Contardi

"... because ***written recipes are usually harder to understand****. Usually, there are harder sections that you can't really see, and by watching someone else, you can be guided step by step."*

"I start by studying the recipe and putting everything I need on the table, to be as organized as possible. ***Then I search for an execution of the recipe on TikTok, and I follow it step by step****."*

Interview 3 - Matteo Cappellari

"I completely messed up that first course because I had put in too much pasta compared to how many of us there were... since there were a lot of us, and ***preparations for many people are always more complicated because you have to keep up with the quantities.****"*

"One thing that I think is important is ***having a clear idea beforehand of what you really need for various preparations****, even those that aren't necessarily complex, but for a lot of people."*

Interview 4 - Giovanni Cranco

*"****Culinary school unfortunately gave me little****... I decided to attend* ***additional courses.****"*

*"****Many mistakes are made in the kitchen; through experiments, I try to improve as much as possible to avoid these occurrences****."*

"... I ***also attended various courses****, connecting cooking to the bar world as well, through different courses and masterclasses attended in the last few years."*

"Yes, often it doesn't turn out well, and many times I don't realize it while I'm working... But then afterwards, let's say from the final result point of view, ***I understand it, and so I decide to improve by experimenting a lot****."*

Interview 5 - Francesco Cranco

"...I'll give you a practical example: ***when I have to make a dish that uses Grouper, but I don't have it available****, ...,* ***I am forced to change****, maybe using Amberjack instead of Grouper."*

*"****The most complicated part is keeping the rhythm, so the timing****. Timing is the fundamental thing..."*

*"Basically, it happened that I put too much salt in the tartare, b****ut I didn't have any other prawns to remake it, so I adjusted it, let's say, by adding some aromatic herbs.****"*

*"****I downloaded a very famous course*** *on my phone called Academia... And from there, lately, especially last winter,* ***I tried making many leavened products and desserts****..."*

"If I have to do new things, one of the first things I consult is an App*, because you find everything there, and it's an App made by Michelin-starred chefs..."*

Interview 6 - Diego Garzia

"Unfortunately, I have to say no [to the question of whether culinary school provided all the tools], ***but the issue is that the school does the maximum it can within its limits****. Because, rightly so, there are many students... So, I* ***recommend more home practice, in addition to school.****"*

"Well, when you want to experiment with a recipe, you definitely start from the raw materials... ***so you also base it on the price.****"*

*"****Online****, I definitely also learned about bread-making, meaning doughs... all the procedures to do it, because to prepare the dough it takes about 2 hours, excluding the rising time which is 24-48 hours."*

"But online it depends on the recipe, because ***if I have to make a more complex dish, I need a tutorial.****"*

"So, a recipe that went wrong was a brownie... ***to prevent this error, one can definitely consult some recipe with a tutorial."***

Interview 7 - Adriana Manzo

*"****The chosen recipe must definitely be very simple, easy, fast, because unfortunately nowadays users' choices are made based on many necessities****... there is a need to have easy, fast, and uncomplicated recipes."*

*"****Even a traditional recipe can be transformed into a modern key****, making it simpler, lightening it with some ingredients, being a bit faster... but yes, nowadays almost any recipe is often 'transformed'."*

"Obviously, even potato peels, instead of being thrown away, ***can be reused, perhaps using the air fryer to make snacks...****"*

"Yes, ***if you organize the kitchen lines*** *(mise en place), preparing all the ingredients, dosing them right at the start,* ***so already having the weight of the various ingredients, you definitely eliminate big, big problems****."*

"If you have the passion, ***then slowly you also refine the technique and therefore the possibility of improving a recipe, of making it yours, by inserting new ingredients, new techniques, and [overcoming] the initial difficulties."***

3.c Synthesis

3.c.i SUMMARY

First, our interviews revealed that for almost all participants, cooking is experienced not only as a daily necessity but also as a creative passion. A common theme found across all respondents is the need for personal organization, such as time management and preparation strategies, alongside the need to test and refine recipes repeatedly while remaining mindful of waste.

Some participants cook based on availability (what they have at home) and may avoid attempting more complex dishes due to the **cost**. Furthermore, another fundamental point is their **approach to mistakes**: specifically, how they try to **prevent** them or **fix** them during the process."

3.c.ii Brainstormed user needs

In order to find as many user needs as possible, we adopted an hybrid approach combining individual analysis and teamwork.

First, each team member reviewed independently the gathered interview material (notes, audio recordings, transcriptions), looking for latent needs emerging from the interviewees' words. Subsequently, during a debriefing phase, we shared our findings and associated potential needs with each participant.

By consolidating these insights, we identified the **following list of needs**, filtering out redundancies and simple reformulations.

Our interviewees need to:

- Make their planning efficient
- Improve their workflow
- Improve waste management
- Learn from mistakes
- Keep track of expiration dates
- Keep costs under control
- Find a way to make up for mistakes
- Have visual clues in order to learn more efficiently
- Deal with the seasonality of the ingredients
- Learn more techniques, unsatisfied with culinary school
- Experiment a lot, even by making mistakes

3.c.iii Deep user needs

We then selected the needs that were common to the majority of the interviews (in some form or another), and that were also among the ones that the domain expert underlined in their answers.

Here we present the list of 4 deep user needs, each associated with the key quotes from the interviews that substantiate our findings.

1. NEED TO MAKE PLANNING EFFICIENT: from the interviews we were able to notice how the preliminary steps of the preparation are crucial. Well-structured planning can really make the difference between a successful cooking session and a disaster.

KEY QUOTES

According to Adriana Manzo:

• *"If one **organizes** the kitchen lines, let's say, by preparing all the ingredients, dosing them already at the beginning... **huge problems are eliminated**. So, the realization of the recipe is sped up, but inconveniences are also avoided."*

According to Andrea Contardi:

• *"First of all, I look at the recipe, I **see what is needed** and put it on the table... so as to be **organized**... and make a consequential choice of things to do."*

According to Diego Garzia:

• *"You start from a mise [en place], which means you **organize everything you need** to make that preparation... for example, if I have to make a sauté I prepare the cutting board, the knife, the bowl for scraps, the bowl for preparations and that's it."*

According to Francesco Cranco:

• *"I **make a list of ingredients**... asking obviously always those who have to taste it... I also regulate myself with **intolerances**."*

• *"The most important thing is always a **clean mise en place**, therefore ingredients already washed, cleaned, ordered... for example, I first prepare the dessert, the sauces... and then I go with the vegetable cuts."*

According to Giovanni Cranco:

• *"At the beginning one **prepares... a general line**... I write in, let's say, a **notebook**... and then the actual preparation begins."*

• *"I focus a lot also on the drawing... to bring, let's say, a realization from the aesthetic point of view."*

According to Matteo Cappellari:

• *"It's important to **understand beforehand** what you really need... not finding yourself halfway through preparation where you need a utensil, a piece of crockery and you can't find it because you didn't prepare it before."*

• *"Once I made a mess because I didn't make a checklist of all the things needed before... **Having everything ready**, in my opinion, is a thing that is **always good to improve**."*

• *"Surely the **choice of the raw material**, in my opinion, is the most fundamental part of all."*

According to Samuele Pagano:

• *"I try to study it as best as possible through various sources... I **try to understand well what I have to do...** buy the right things so as not to waste either food or money."*

• *"Check first all the labels [for expiration dates] ... **doing it before** prevents the dish from being ruined."*

• *"If, for example, there is a person... intolerant to lactose... I try to avoid ingredients containing lactose... the kitchen is never an end in itself, but you are doing it for someone."*

2. NEED TO IMPROVE WORKFLOW: once the stoves are hot, chaos management gets crucial according to our interviewees. Specifically, when you are inexperienced, keeping the rhythm while cooking and cleaning simultaneously is one of the main concerns.

KEY QUOTES

According to Andrea Contardi:

• *"When I cook, I make a real **mess**: I get everything dirty, so perhaps I could use a bit more cleanliness..."*

According to Francesco Cranco:

• *"The most complicated part, let's say, is **keeping the rhythm**: timing is the fundamental thing."*

• *"One thing I absolutely must change is that when I start cleaning the vegetables, maybe I won't throw away the scraps immediately... I need to learn to immediately throw them away, so that with the order on the counter there is also mental order."*

According to Samuele Pagano:

• *"It is important to **be organized while you cook**... you must take a thing, use it, put it immediately back in place... you need to do both things simultaneously, cooking and cleaning."*

• *"Initially I remember that the **main difficulties were the times and the cleanliness**, that is the disorder... I didn't know how to regulate times initially... and then afterwards the kitchen was a disaster."*

3. NEED TO IMPROVE WASTE MANAGEMENT: The interviews reveal that waste management is a central concern for all interviewees. For less experienced cooks, the primary strategy is prevention: they focus on careful planning, buying only what is necessary, and avoiding execution errors to ensure the food remains edible.

KEY QUOTES

According to Adriana Manzo:

• *"We try to **eliminate waste**... for example, even the potato peel... instead of being thrown away, can be reused to make snacks."*

• *"Scraps can be used to make another dessert... generally we **try to exploit** as much of the same ingredient."*

According to Andrea Contardi:

• *"...even if it doesn't come out perfect, I eat it to avoid wasting food, because **I don't like to waste**."*

According to Diego Garzia:

• *"If I have **some food left over**, I try to create a dish with it."*

• *"The [onion] peel is a quite good part... it can be added to the dough of a gnocchi to make pink gnocchi, and in the end, **you haven't thrown away anything** of the onion."*

• *"When we filet... a carp... the bone, the head, the tail we throw them away, but we can easily make a broth."*

According to Francesco Cranco:

• *"Speaking of scraps with vegetables, the easiest things are broths: you take all the scraps and make a vegetable broth."*

• *"A carrot peel, for example, washed, dried, then you can fry it... you can make an oil with basil stems..."*

• *"With fish scraps obviously you can make a fumet, same thing goes for meat, trimmings... they make the base of a stock."*

According to Giovanni Cranco:

• *"The potato peels, for example, I fry them... scraps can be used for infusions, broths, and so I try to waste as little as possible."*

According to Matteo Cappellari:

• *"Every time I make fish, I almost always make a bisque with the heads. All the carapaces which are a fundamental flavor."*

• *"It was amazing how many dishes revolved around a single ingredient of which, however,* ***all parts are used****... at 100%."*

According to Samuele Pagano:

• *"I try to understand well what I have to do... buy the right things to* ***not waste food and money****."*

• *"It happened that... I had forgotten something in the oven... in that case, unfortunately... you can't do anything but throw the dish away. And it is an absurd shame because maybe you wasted many ingredients."*

4. NEED TO LEARN FROM MISTAKES: the interviewees highlighted how the "mistake" is a crucial (and inevitable) step along the culinary growth path. The lack of experience often leads to practical issues, such as messing up with doses or timings. According to the following quotes, a *trial and error* based approach is the key to get better.

KEY QUOTES

According to Adriana Manzo:

• *"Obviously there are* ***errors****, you can even throw everything away when you make a recipe... but if you have passion, then* ***slowly you also refine the technique*** *and therefore the possibility of* ***improving*** *a recipe."*

• *"****You have to do****; by doing you* ***make mistakes****, and* ***by making mistakes you acquire experience****, and everything eventually becomes much simpler."*

According to Andrea Contardi:

• *"****I try it, I try it again,*** *I try it again* ***until it comes out really good****."*

• *"That is dictated entirely by the **inexperience** of when one is young... you exaggerate with the doses, that's always how it ends up."*

According to Francesco Cranco:

• *"It happened that I put too much salt in the tartare... so I fixed it, let's say, by adding some aromatic herbs and lemon and, let's say, in the end I saved it. It was **inexperience**... It was my first menu... I had a bit of anxiety... **now I can understand an error of that kind**."*

According to Giovanni Cranco:

• *"I do **several experiments** to bring the dish I have in mind to maximum perfection [...] and **through experiments I try to improve as much as possible**."*

According to Samuele Pagano:

• *"I didn't know how to manage cooking times initially when I started... But then slowly, anyway **with experience, I learned** to manage times better."*

• *"For the problem of time, in my opinion, **the solution is experience, cooking a lot** and therefore understanding the cooking times well."*

3.d Solutions

3.d.i DEEP USER NEEDS SOLUTIONS

To come up with the possible ways to address the deep user needs, we opted for a cooperative strategy. Each member had to propose at least two possible solutions, while others had to approve them.

By iterating this process over all the needs, we gathered the following solutions.

- **Need to make planning efficient**:
    - write down a checklist of all the needed ingredients, doses and utensils
    - sketch a "preparation timeline" to follow
    - look for online video recipes to have a clearer idea of what to do
    - write a ToDo list with all the preparation steps
    - keep track of the guest's intolerances and dietary restrictions

- **Need to improve workflow**:
    - ask somebody to assist you during the preparation
    - divide semantically the working zone (cutting–zone, cooking-zone, waste-zone, etc.)

    - make sure that the instruments are physically available when needed
    - Adopt a "clean as you go" strategy to maintain mental order
    - Synchronize cooking times to handle multiple preparations simultaneously

- **Need to improve waste management**:
    - learn new recipes to use *wastes* (e.g., peels, fish bones, etc.)
    - only buy exactly what you need and nothing more
    - learn new cooking techniques (e.g., drying, infusion, freezing)
    - a calendar-based *expirations reminder*
    - repurpose failed preparations into different dishes instead of throwing them away

- **Need to learn from mistakes**:
    - take note of mistakes as you make them
    - record yourself while cooking
    - practice the same recipe until you make it
    - study the theory on cooking books
    - learn adjustment techniques to "save" a dish while cooking

3.d.ii FINAL SOLUTION

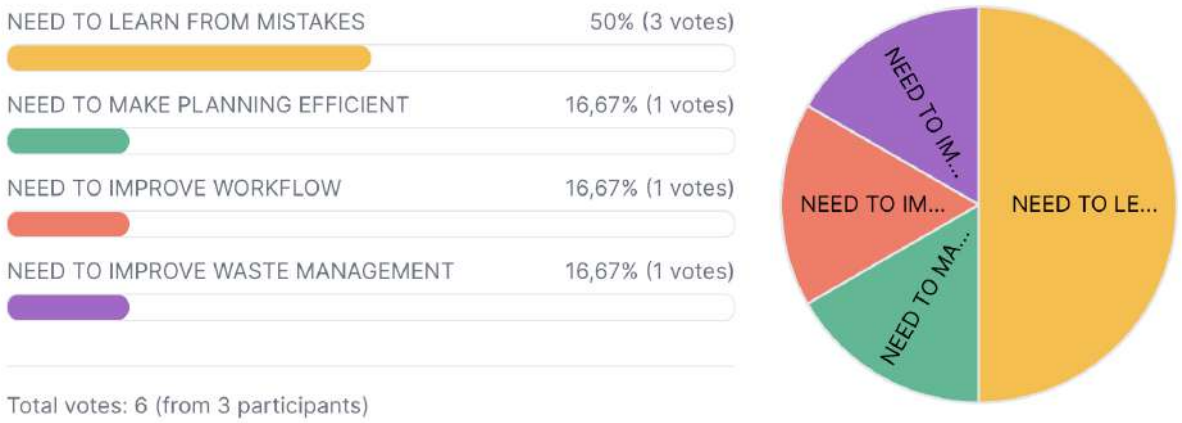


Figure 1 - Need choice

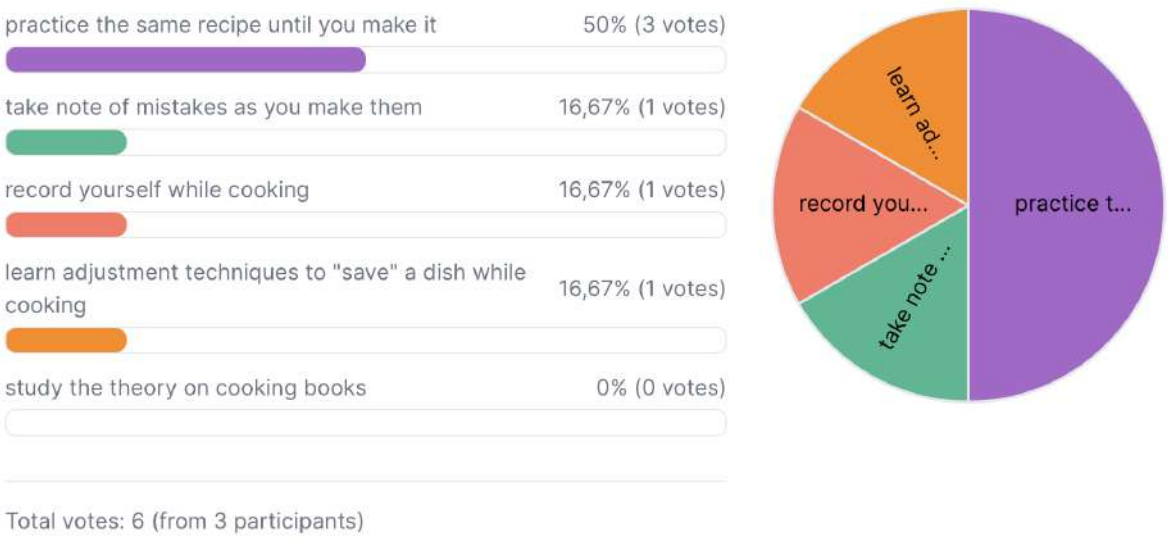


Figure 2 - Solution choice

https://strawpoll.com/

Eventually we discussed which, among the gathered solutions, could have allowed us to create a meaningful and useful product, keeping in mind what the interviewees explained to us.

After the discussion, first we voted our preferred deep user need (fig. 1), then our preferred solution (fig. 2).

As shown by the images, the winning solution was "**practice the same recipe until you make it**". As far as we are concerned, this is particularly effective since we enforce a **learn-by-doing approach** to prevent the user from making the same mistakes repeatedly.

4. Tasks and Storyboard

4.1 Tasks

- Execute an exercise *(simple)*
- Reporting errors in exercises *(moderate)*
- Add ingredients for exercises to your Amazon cart *(complex)*

To design our solution, we tried to think like the potential average user of our application and how the flow of the app could look starting from a situation of need, showing the possible steps the user should go through. We didn't focus on how to complete the task.

After a brainstorming session on which goals would be suitable for our application, we chose the following:

SIMPLE TASK: Execute an exercise

The user starts from a situation where they need to practice something, for example, after trying to cook a dish for real and making several mistakes, the user uses the application to practice and learn from their mistakes.

MODERATE TASK: Reporting errors in exercises

We envisioned our solution as an application that uses artificial intelligence to generate recipes. Since AI might make mistakes, we allow the user to report errors, which will be considered for the generation of future recipes.

COMPLEX TASK: Add ingredients for exercises to your Amazon cart

Eventually, we thought that after practicing with our application, the user might feel ready to try cooking for real. The application allows the user to select the ingredients they practiced with and add them directly to the Amazon cart from the app.

4.b Storyboard

Figure 3 - Storyboard

The storyboard shows the first two tasks, since they are the most effective for demonstration purposes, and the most meaningful for our solution.

We made sure to avoid implementation details, such as the way exercises are handled, to just focus on the goal.
The storyboard shows how the user, after a cooking mistake, desires to learn how to overcome that issue. He decides to let the system generate an exercise that could let him practice a similar scenario to the one he failed in real life (simple task). After completing the exercise, the user knows how to handle the recipe, reaching the goal.

The last panel shows the moderate task, with the user finding an error in the proposed exercise, reporting it in the application.

The main strengths of the storyboard can be found in the immediateness of the representation of the two tasks, and in the generalization of their implementation, leaving a lot of room for the actual prototype.

The weaknesses can probably be found in the depiction of the mobile application case, which might be difficult to translate to other modalities.

5. Low-fidelity Prototypes

5.a Explored Modalities

Among the possible modalities for our application, we decided to explore two in particular: **Smartphone Application** and **AR Application**.

The smartphone has been chosen since it's the most immediate way to do an exercise with a digital tool: it can allow a great variety of exercise modalities and can be used more easily by a wider audience.

The AR headset has been chosen since it can be the only device capable of providing users with digital ingredients, allowing them to try exercises without any waste.

5.b Paper Prototypes

Once we decided the modalities, we designed the core of the two possible versions of the application, adapting the designs to the capabilities and limitations of the two systems.

Mobile Prototype

The Mobile Prototype has been drawn using standard design elements and conventions for mobile applications, leveraging the possibility of having longer screens that can be explored by the user.

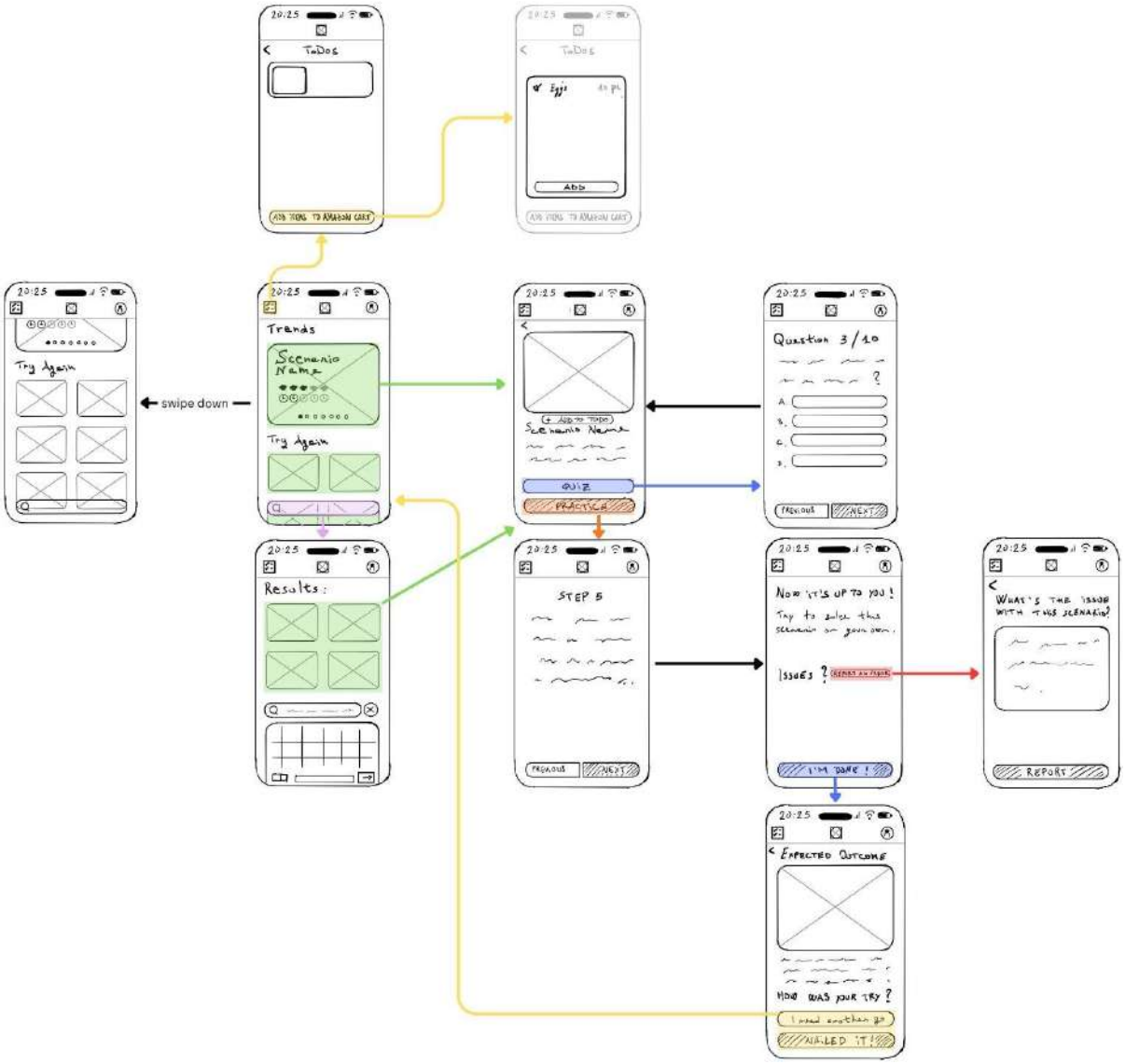


Figure 4 – Mobile Prototype Flow Diagram

The **simple task**, being performing the exercise, is divided in two separate activities, and users can freely decide how to approach them: there are "quiz" sections, where the user has to choose the right option when asked about some parts of the exercise, and "practice" sections, where the user has to follow certain steps in his kitchen, with real ingredients, to successfully complete the exercise. The application then shows the expected outcome, that the user can compare with the result he obtained.

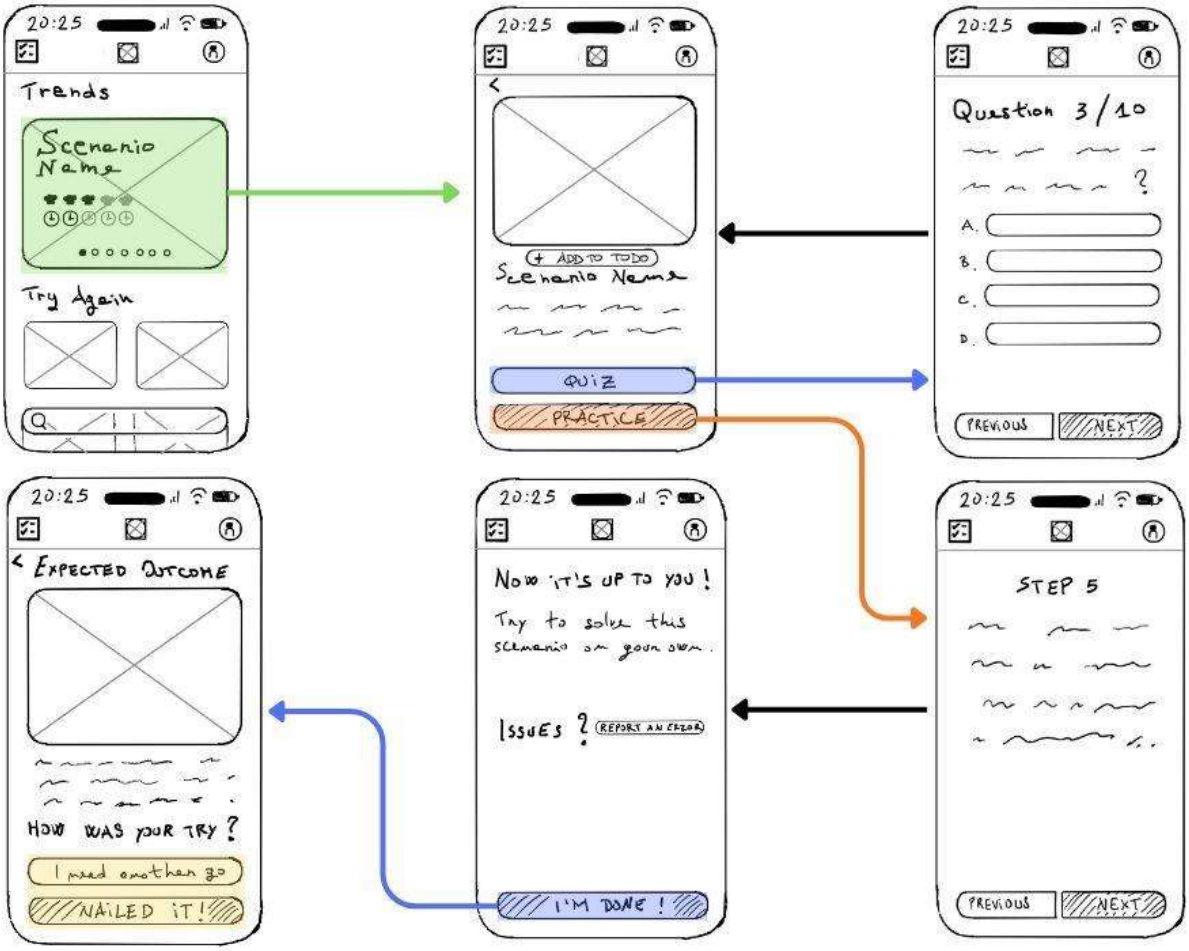


Figure 5 – Mobile Prototype Simple Task Flow Diagram

The "quiz" section allows the user to study the recipe, to approach the "practice" section with the appropriate preparation.

The **moderate task**, being the error report, can start immediately after completing an exercise:

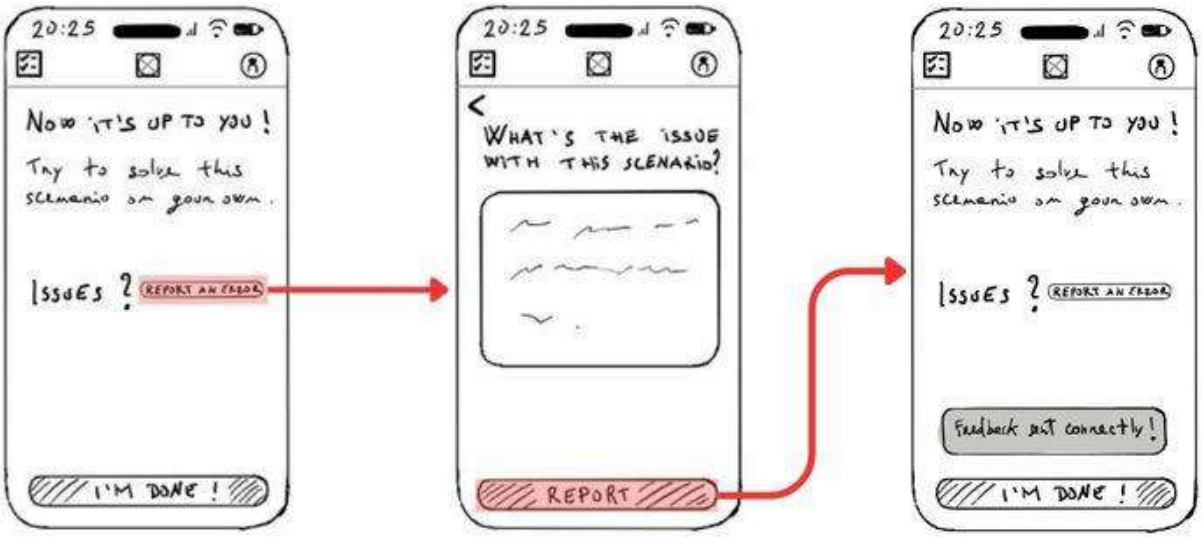


Figure 6 – Mobile Prototype Moderate Task Flow Diagram

The **complex task**, being adding ingredients to the Amazon cart, has been designed to allow users to choose to add to the amazon cart only certain ingredients

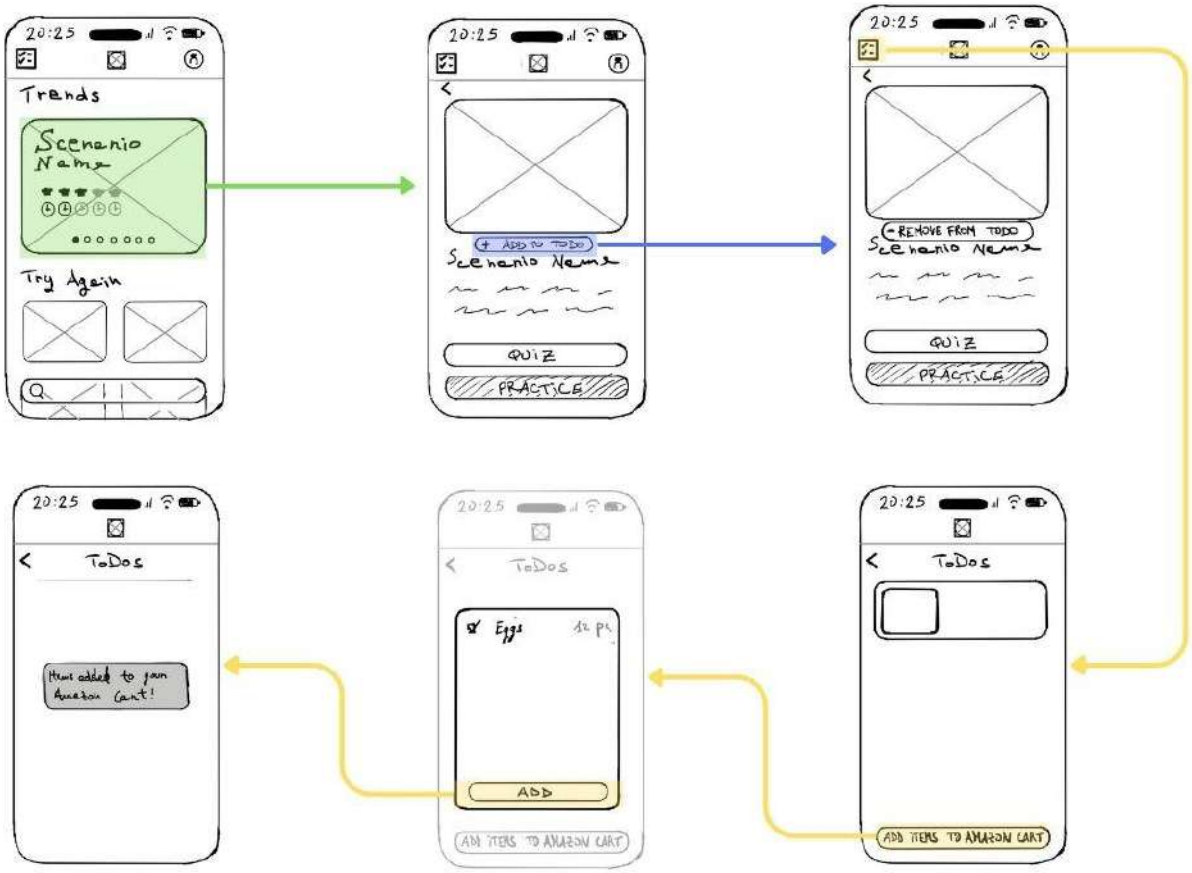


Figure 7 – Mobile Prototype Complex Task Flow Diagram

AR Prototype

The AR Prototype has been drawn considering a basic kitchen setting in the background. In this version, the user can perform exercises using virtual ingredients, while guided by the app and being asked some theory questions to prevent mistakes.
The interface is less cluttered than the mobile one, to avoid the information overhead a user might experience while wearing a headset.

The flow chart for the full prototype is the following:

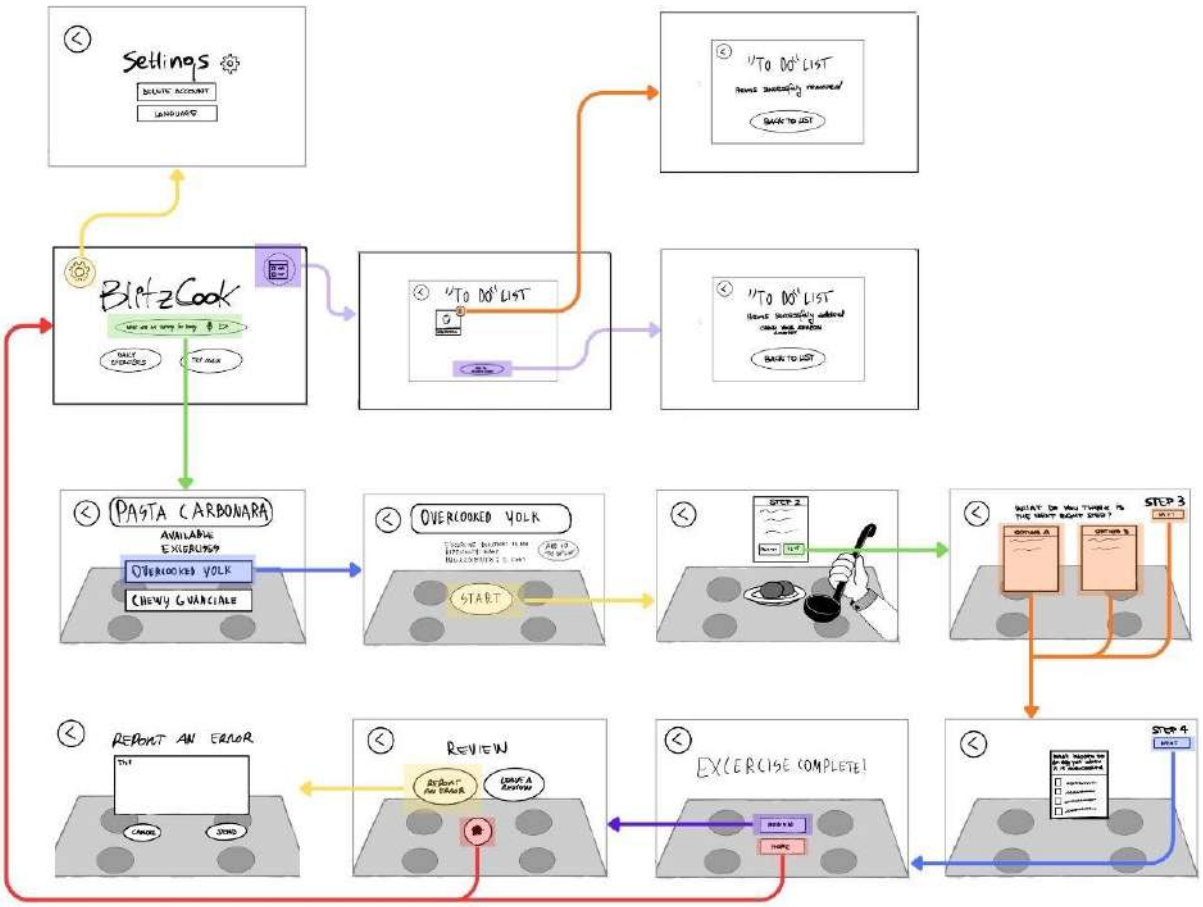


Figure 8 – AR Prototype Flow Diagram

The **simple task**, being performing the exercise, now involves real time participation of the user:

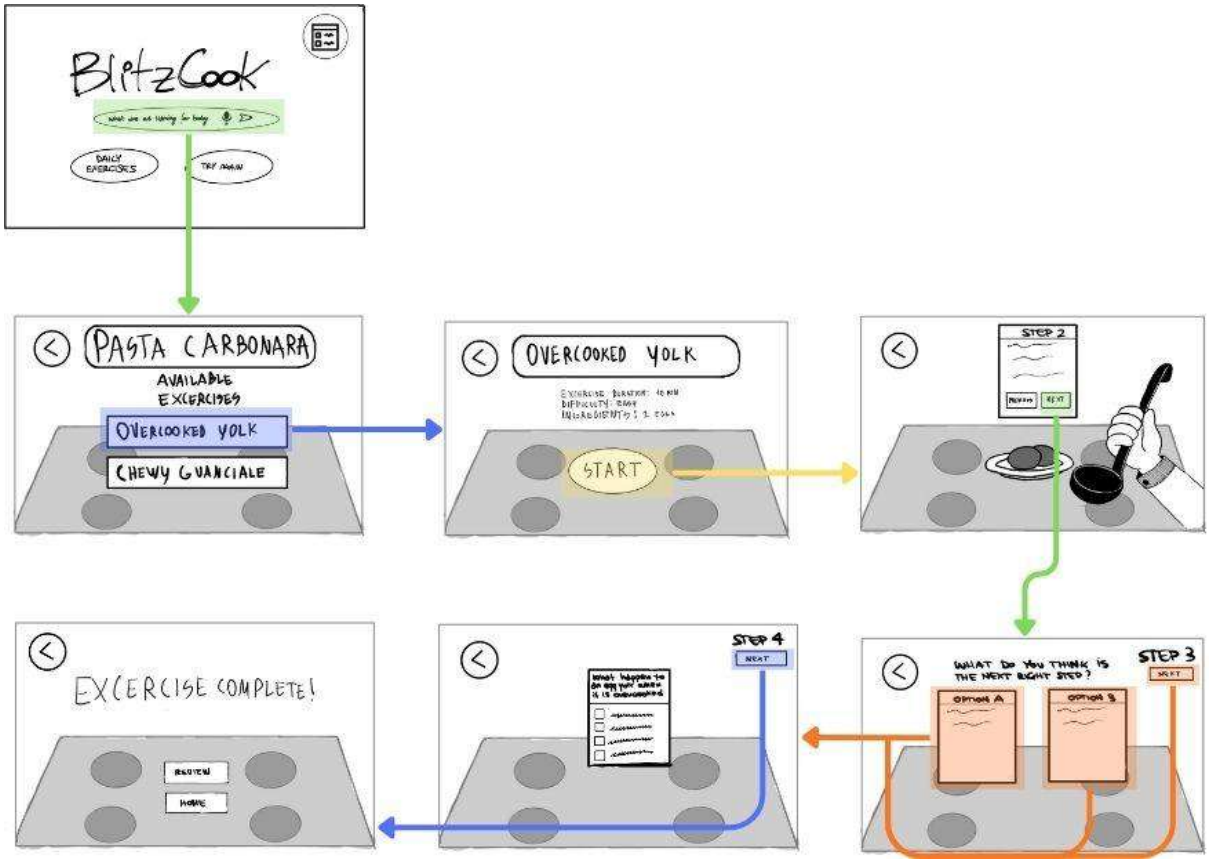


Figure 9 – AR Prototype Simple Task Flow Diagram

Like in the storyboard, users can train for a recipe as many times as they need, until they are finally able to approach it without any issue.

The **moderate task**, being the error report, can start immediately after completing an exercise:

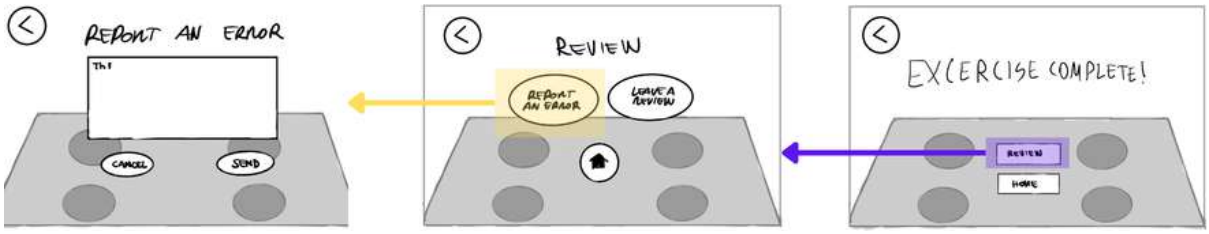


Figure 10 – AR Prototype Moderate Task Flow Diagram

The **complex task**, being adding ingredients to the Amazon cart, now has a much simpler flow, since adding too many options can clutter the AR space:

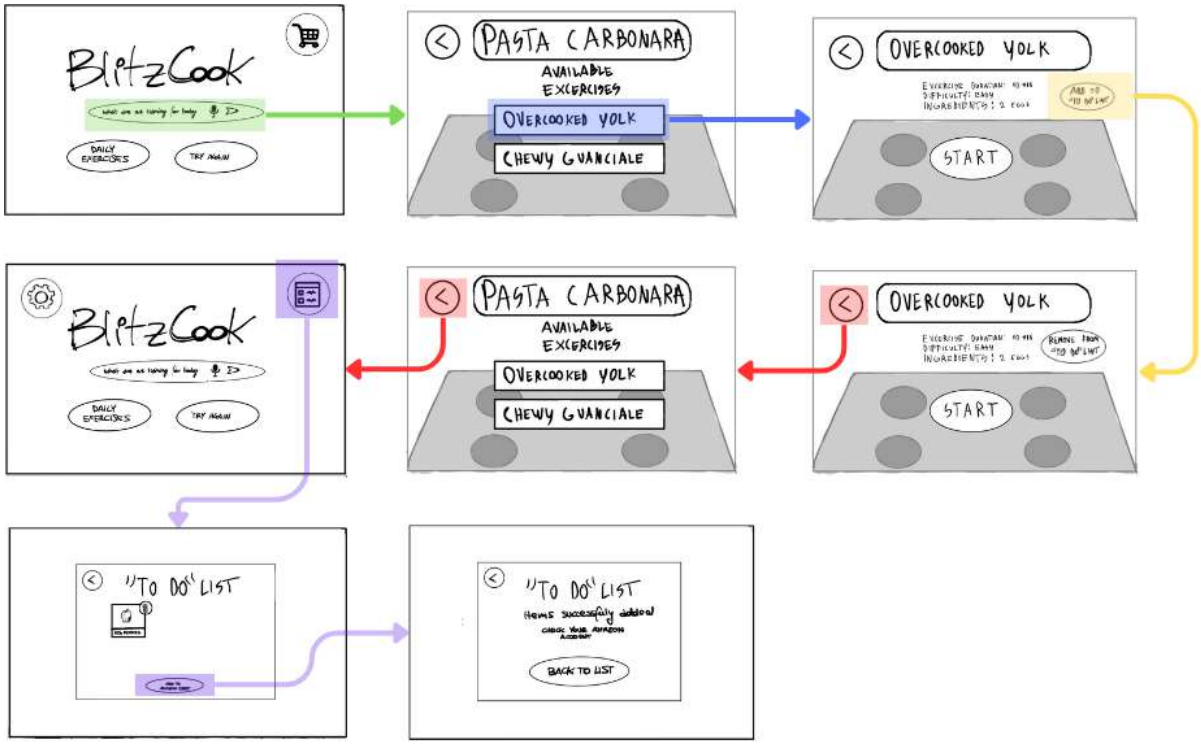


Figure 11 – AR Prototype Complex Task Flow Diagram

5.c Heuristic Evaluations

5.c.i Preparation

Starting from the previously drawn paper prototypes, we prepared printable sheets containing the various screens, that we then printed on paper sheets. We also prepared additional sheets containing the various possible options for buttons, screens and text:

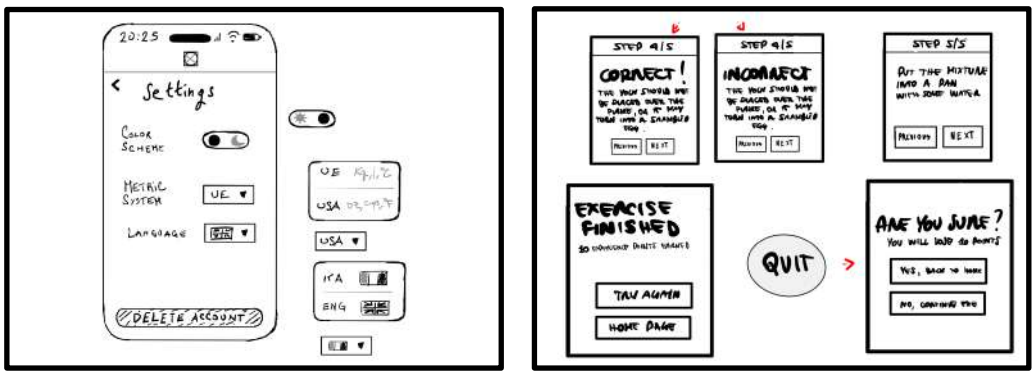


Figure 12 – Sample prints

For the mobile prototype we prepared a paper smartphone, in which we slid the various screens, where we laid the different status of buttons and windows when needed.

Figure 14 – Paper smartphone

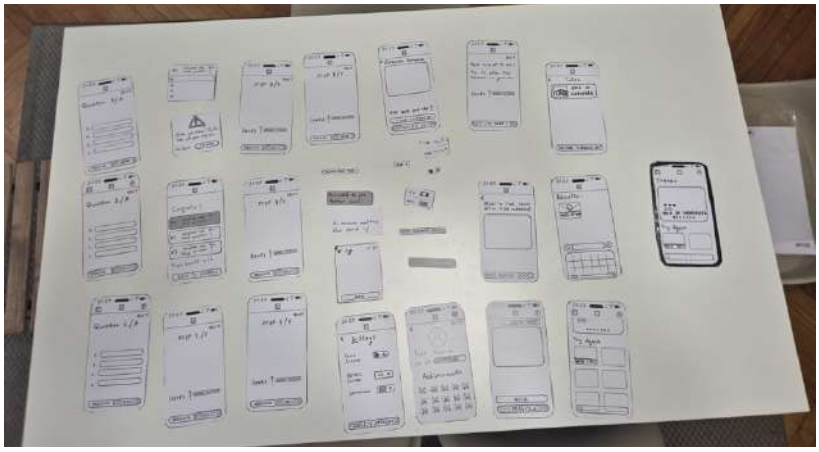


Figure 15 –Printed Mobile prototype

For the AR prototype we prepared full sized prints for the menus, such as the main menu and the recipe selection. The exercise screen has instead been handled by putting the "steps" windows and the buttons on a blank sheet of paper, since the AR exercise can have anything as a background.

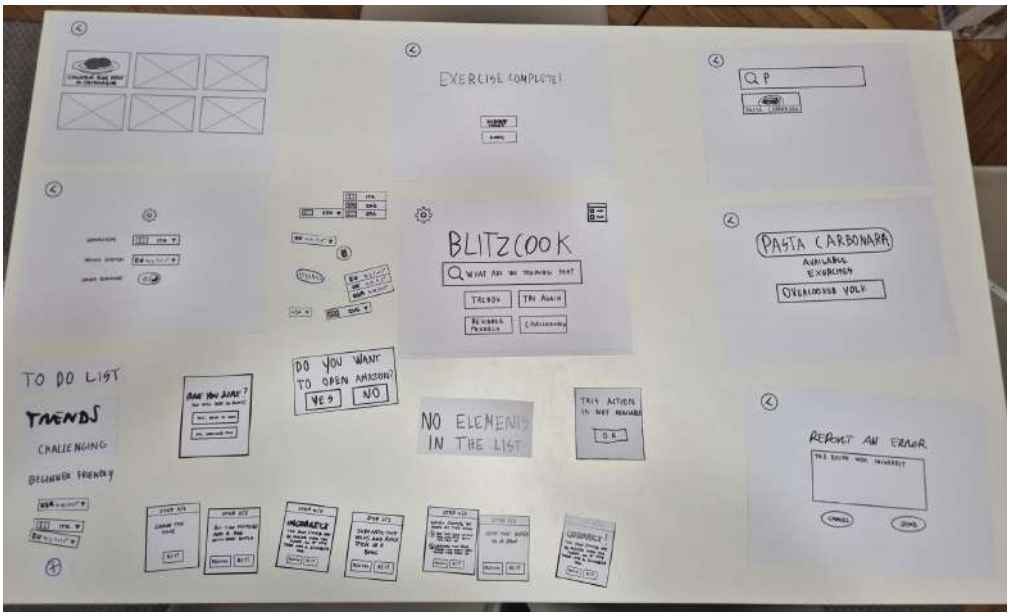


Figure 16 – Printed AR prototype

While handling such many small pieces of paper during the evaluations was challenging, it resulted in a very robust prototype, enabling evaluators to fully explore the applications without incurring in non-planned situations.

5.c.ii Execution

The team showed the prototypes with a member acting as the computer, moving the screens accordingly, and another explaining the project's domain and functionalities, describing the three tasks and answering any questions, making sure that the evaluators had all the information required to test the application appropriately, but still paying attention on not providing any hint to the evaluator.

While evaluators were able to successfully interact with any component of the prototypes, they still found some violations.
The less severe violations focus on consistency, mainly involving button shapes or the inconsistent usage of text and icons.
The more severe ones for the Mobile prototype involved the To Do List, which has been considered prone to user errors, and had some flaws involving missing information.
The more severe ones for the AR prototype involved the "Quiz" and the "Steps" popups in the exercises, which have been considered too small and confusing for the purpose of the exercise, and the "To Do List", where the evaluators believed that the simplified interface left too little options to the users.

5.c.iii Violations found

For the Mobile application, the violations are the following:

H1 — Visibility of System Status

1. **WHERE:** Amazon Cart Flow – After clicking "Add" in the ingredient popup
   WHAT: Popup "Items added to your Amazon Cart!" appears, but items disappear from To-Dos and the cart state is not visible in-app
   WHY: Users cannot verify what was added or the cart contents without leaving the app
   SEVERITY: 3
2. **WHERE:** Profile Screen / Dashboard
   WHAT: "Edit Profile" button is greyed out and non-clickable
   WHY: Users cannot understand whether the feature is locked or not implemented
   SEVERITY: 1
3. **WHERE:** ToDos page
   WHAT: No indication of whether practice/quiz in a scenario was already completed
   WHY: Users cannot track their progress
   SEVERITY: 2
4. **WHERE:** Reporting a problem in a scenario
   WHAT: No feedback or monitoring of a reported issue
   WHY: Users cannot know whether the issue was addressed, reducing clarity and trust
   SEVERITY: 2

H3 — User Control and Freedom

5. **WHERE:** Amazon Cart Flow – "Items added" confirmation
   WHAT: No "Undo" option after adding items
   WHY: Mistakes cannot be reversed without switching to the Amazon app
   SEVERITY: 3

6. **WHERE:** ToDos – Add to Amazon Cart
   WHAT: Items are added to the cart instantly on tap, without confirmation
   WHY: This may cause accidental additions; reverting requires switching to Amazon
   SEVERITY: 3

H4 — Consistency and Standards

7. **WHERE:** Practice Flow – "Expected Outcome" screen
   WHAT: Top-left back arrow restarts the exercise instead of navigating back
   WHY: Breaks standard navigation conventions and traps the user
   SEVERITY: 3
8. **WHERE:** Dashboard – Top Navigation Bar
   WHAT: Central icon appears clickable but is not
   WHY: Users expect a central navigation element to perform an action
   SEVERITY: 2
9. **WHERE:** Reporting a problem in a scenario
   WHAT: Users can continue the scenario after reporting an issue
   WHY: A faulty step may invalidate later steps; expected behavior would be interruption
   SEVERITY: 2
10. **WHERE:** Settings – "Specify your language"
    WHAT: Selected language shows a name; menu shows only flags
    WHY: Inconsistent notation and unclear for users who don't recognize flags
    SEVERITY: 2
11. **WHERE:** End page of a scenario
    WHAT: App says "now it's your turn in the real world!", but scenario is removed from ToDos
    WHY: Removal prevents users from revisiting steps/ingredients
    SEVERITY: 2
12. **WHERE:** Main page – Search bar
    WHAT: Back button uses an "X" instead of an arrow, unlike the rest of the app
    WHY: Inconsistent visual language
    SEVERITY: 1
13. **WHERE:** Scenario practice steps
    WHAT: "Quit" button appears in the upper-right corner
    WHY: Cancel/undo actions are conventionally on the left
    SEVERITY: 1

H5 — Error Prevention

14. **WHERE:** End of a practice scenario
    WHAT: "Quit" remains visible while an "End" button appears in a new location
    WHY: Could mislead users into discarding progress instead of saving it
    SEVERITY: 3

15. **WHERE:** Settings – "Specify your language" (Overlap: H4 + H5)
    WHAT: Flags without names may cause users to pick the wrong language
    WHY: Increases likelihood of selection errors
    SEVERITY: 2

H6 — Recognition Rather Than Recall

16. **WHERE:** ToDos – Ingredient list
    WHAT: No indication of which scenario each ingredient belongs to
    WHY: Users must reopen scenarios to recall required ingredients
    SEVERITY: 3

H9 — Help Users Recognize, Diagnose, and Recover from Errors

17. **WHERE:** Search Results screen
    WHAT: "No scenario matches this search :/" without suggestions
    WHY: No help is provided to recover or refine the search
    SEVERITY: 1
18. **WHERE:** Reporting a problem in a scenario
    WHAT: After reporting, users can quit or proceed but receive no explanation
    WHY: No guidance on why the error occurred or how to resolve it
    SEVERITY: 3

For the AR application, the violations are the following:

H1 — Visibility of System Status

1. **WHERE:** To-Do List Page
   WHAT: Delete exercise buttons are too large and overlap exercise names.
   WHY: The user cannot see what they are deleting, reducing clarity and increasing error likelihood.
   SEVERITY: 3
2. **WHERE:** Recipe Page – Add/Remove from To-Do List Button
   WHAT: Only the button label changes when a recipe is added/removed.
   WHY: Change is too subtle; no additional feedback confirming the action.
   SEVERITY: 2

H2 — Match Between System and the Real World

3. **WHERE:** To-Do List Page
   WHAT: "Order Ingredients" button is misplaced: it appears in the general To-Do List, but no specific recipe can be selected.
   WHY: Ingredients are purchased per recipe, not globally. The system does not match real-world workflow.
   SEVERITY: 4

4. **WHERE:** Home Page – "Try Again" Button
   WHAT: The button opens a list of completed exercises, not a retry of the last failed task.
   WHY: Label suggests immediate retry; behavior suggests browsing past exercises.
   SEVERITY: 2
5. **WHERE:** Quiz Page
   WHAT: Users must click a small numbered button to answer, but the layout resembles a simple list.
   WHY: Users expect clearer affordances for selecting answers; poor layout hides intended interaction.
   SEVERITY: 3

H3 — User Control and Freedom

6. **WHERE:** Search Exercise Page
   WHAT: No button to clear typed text in the search bar.
   WHY: Users cannot easily correct mistakes.
   SEVERITY: 3
7. **WHERE:** During the Exercise
   WHAT: Steps can only be moved, not hidden.
   WHY: In AR, visible floating windows may obstruct reality; users need a toggle.
   SEVERITY: 2
8. **WHERE:** "Order Ingredients" Global Action
   WHAT: The button orders *all* recipes' ingredients without selection.
   WHY: Users cannot choose specific recipes; high risk of unintended purchases.
   SEVERITY: 4
9. **WHERE:** Voice Input Fields
   WHAT: Users can add voice input but cannot edit or delete it.
   WHY: Zero edit control; only workaround is leaving and re-entering the page.
   SEVERITY: 3

H4 — Consistency and Standards

10. **WHERE:** Step-by-Step Page
    WHAT: "Next" button is square while other buttons are round.
    WHY: Breaks visual consistency.
    SEVERITY: 1
11. **WHERE:** During the Exercise
    WHAT: "Quit" is written as text instead of using the iconography used elsewhere.
    WHY: Inconsistent navigation symbols across the app.
    SEVERITY: 2
12. **WHERE:** Home Page – Search Bar
    WHAT: Search bar behaves like a navigation trigger, not an input field.
    WHY: Violates standard search bar behavior and UI expectations.
    SEVERITY: 2

13. **WHERE:** Filter Exercises Page – Search Bar (Overlap: H6 + H4)
    WHAT: No visual indication that voice input is required.
    WHY: Users must remember interaction rules not visible on screen.
    SEVERITY: 2
14. **WHERE:** To-Do List Page & Buttons
    WHAT: Use of the label "To-Do List" instead of common "Favorites" / heart icon.
    WHY: Diverges from widespread app conventions, increasing cognitive load.
    SEVERITY: 2
15. **WHERE:** End of Exercise Page
    WHAT: Buttons are rectangular instead of the app's usual ellipsoidal style.
    WHY: Inconsistent button styling weakens interface coherence.
    SEVERITY: 1

H5 — Error Prevention

16. **WHERE:** Quiz Step Page
    WHAT: "Next" button is active even before answering.
    WHY: Users can skip questions, undermining learning objectives.
    SEVERITY: 3
17. **WHERE:** Quiz Step Page
    WHAT: Answer buttons ("A", "B") are too small.
    WHY: Small tap targets increase misclicks and frustration.
    SEVERITY: 3
18. **WHERE:** End of Exercise – "Next" Button
    WHAT: Button immediately ends or moves forward without confirmation.
    WHY: Users may accidentally finish the exercise prematurely.
    SEVERITY: 3
19. **WHERE:** To-Do List – Individual Recipe Removal
    WHAT: Removal happens immediately without confirmation.
    WHY: Users may accidentally delete items, especially with gaze-based input.
    SEVERITY: 3

H6 — Recognition Rather Than Recall

20. **WHERE:** Start of the Exercise
    WHAT: Recipe text does not match the exercise or is missing.
    WHY: The user must recall which recipe was selected instead of recognizing it.
    SEVERITY: 2

H7 — Flexibility and Efficiency of Use

21. **WHERE:** During the Exercise, Quiz Step Page
    WHAT: Shortcut to skip quizzes is not correctly handled.
    WHY: Experienced users are slowed down unnecessarily.
    SEVERITY: 2

H8 — Aesthetic and Minimalist Design

22. **WHERE:** Start of the Exercise
    WHAT: Book-style design cannot display long ingredient lists properly.
    WHY: Small space reduces readability for long lists.
    SEVERITY: 2
23. **WHERE:** Report an Error Page
    WHAT: Cancel button duplicates the back button.
    WHY: Redundant controls clutter the interface and cause confusion.
    SEVERITY: 2
24. **WHERE:** To-Do List & Recipe Page
    WHAT: Two different removal methods (corner "X" + page button).
    WHY: Adds unnecessary complexity and visual clutter, especially with gaze input.
    SEVERITY: 1
25. **WHERE:** Exercise Page – Information Overlay
    WHAT: Info window is small, not resizable or expandable.
    WHY: Reduced readability; no support for users needing larger text.
    SEVERITY: 2

5.d Selection

The group eventually decided to pick the **AR prototype** as the one that will be developed further, mainly for three reasons:

- It provides a more unique and engaging user experience compared to the mobile application. The AR prototype leverages augmented reality technology to create an immersive environment, which is more aligned with the project's goals.
- An AR application involving virtual cooking exercises, with virtual ingredients, does not currently exist, with ours being the first application of its kind.
- As junior developers, we found the challenge of working with AR technology way more appealing, since it constitutes an opportunity to learn and grow our skills in a cutting-edge field.

Moved Features

After making this choice, the group decided to implement a feature from the Mobile prototype: a "select ingredients" screen where the user can choose which ingredients to order between the ones in the To-Do list. This was initially removed from the AR version because it felt too "complex" for an AR app, but the heuristic evaluation has shown that it was necessary.

6. Medium to High-Fidelity Prototype

6.a-b-c Tools & Screens

Our prototype has been made in **Figma**, since it's the tool the team had the most familiarity with.

Prototype link: https://www.figma.com/design/SSdVrlBWPV7SVXnu10uJJK/AR-Prototype?node-id=0-1&t=OVEm5aXhT6QNFqjL-1

Our Mid-Fi prototype includes the following screens:

- **Selection Screen**: here the user, after having successfully typed his prompt, can choose between different cooking exercises. Once chosen, the user can retrieve all the general information about the selected exercise (such as ingredients, difficulty, and estimated time), add it to ToDos, and start it.

- **Cooking Screen**: this is where the user can practice trying to solve the cooking exercise. It's possible to show a hint for the current exercise, go back to the selection screen, and complete it (by clicking on the screen). While the exercise itself was impossible to implement due to technical limitations, this screen helped us finalize the design of the “core” part of the application.

Moreover, the group has rediscussed the concept of "step" during the exercise (Cooking Screen), and we decided to remove it completely from the AR prototype. The user will now be able to retrieve all needed information about the recipe in the Exercise Screen, and during the exercise, he will have to rely on his skills to complete it. In case he needs help, we've added a "Hint" button that will provide useful tips about the recipe.

Additionally, the "theoretical quizzes" have been removed, making the exercises completely practical.

Lastly, since the user now is completely free to complete the exercise on his own, we decided to add a “constraint” in the form of a timer.

6.d Violations Handling Plan

After analyzing the violations found in the AR Low-Fi prototype, we fixed the following ones for the Mid-Fi prototype:

- Violation #2: A confirmation popup has been added when adding/removing a recipe from the To-Do list, to provide clear feedback to the user about the action taken.
- Violation #5: Since theoretical quizzes have been completely removed, the buttons have been removed as well.

- Violation #7: Even though the "steps" concept has been removed, a "Hint" button has been added during the exercise, that opens a small overlay that can be closed at any time.
- Violation #14: The button has been replaced by an icon, ensuring consistency.
- Violation #16: Same as violation 5.
- Violation #17: Same as violation 5.
- Violation #18: Same as violation 5.
- Violation #20: The exercise title is now shown, accompanied by further information about what it includes.
- Violation #21: Same as violation 5.
- Violation #22: To improve readability and better accommodate long lists, both the Ingredient List and Description layouts have been redesigned, by making them scrollable.
- Violation #23: The additional "back" button has been removed.

Alongside these already applied fixes, we’ve planned to solve the remaining violations in the following way:

- Violation #1: The “delete” button will be moved at the top of the screen, enhancing interface clarity and reducing error likelihood.
- Violation #3: The possibility to select individual ingredients like in the mobile prototype will be added, but ingredients will now be shown without being grouped by recipe to reduce visual clutter.
- Violation #4: The "Try Again" category will be renamed to make clear that it contains all previously completed exercises.
- Violation #6: We will implement a way to repeat the input as many times as needed and confirm it separately.
- Violation #8: Same fix as in violation 3, as they refer to the same screen.
- Violation #9: Same as violation 6. A way to directly edit the input without repeating it will not be implemented since it would require a very complex and cluttered UI.
- Violation #10: Every button's shape will be unified.
- Violation #12: Same as violation 6.
- Violation #13: Same as violation 9.
- Violation #15: Same as violation 10.
- Violation #19: The “delete” button will be moved at the top of the screen, far from any other button, and will be one affecting all of the selected ingredients. A pop-up asking for confirmation will not be added since the team believes that adding another step for confirming the action using Gaze Input can become too time consuming, and moving the button to a different location is sufficient to avoid accidental “presses”.
- Violation #24: Same as violation 19.
- Violation #25: The “description” section in the recipe will allow “expansion”, enabling users to view it in a larger font.

After careful consideration, the team has decided to **dismiss** Violation #11, as it does not agree that it represents a valid issue. The reason is that the "Quit" button needs to be differentiated, since it's the only way to make the user exit the exercise. Also, every button that performs an important action in the app (such as starting the exercise) contains text as well.

7. High-Fidelity Prototype

7.a Technology

Since AR applications were an unexplored field for all the members in the group, we decided to limit the impact by using a **Viro**, a **React Native** library for AR and VR applications. This choice simplified the work a lot, since every member knew React from the Web Applications I course. The library also supported the handling of 3D models, image tracking and particles, providing components for everything we needed.

For data persistence, we opted for **SQLite3**, a lightweight database with which we already had experience.

For microphone support, we decided to use **React Native Voice**, a simple yet effective library for vocal input.

7.b Repository

https://github.com/polito-hci-2025/blitzcook

7.c Design Choices and Changes

We decided that the best way to interact with the application is through **Gaze Input**, which enables users to select certain options by pointing a pointer to them and waiting a couple of seconds. This approach enables users to interact with the application hands free, which we believed was very important for the purpose of the application, since users will have their hands busy while performing the exercises.
While this solution is not natively supported by Viro, we were still able to implement it using its native components.

For what concerns the text input, we believed that the best way for the target device was through **Vocal Input**, for the same reasons listed above. While not as accurate as a keyboard input, or even a “virtual keyboard” with buttons pressed with the Gaze Pointer, this solution has been deemed appropriate for the purpose of the application, since users will probably use the app in their house, which is a reasonably quiet environment.

For the ingredient's simulation, we decided to use printable **codes** depicting the various ingredients and tools, enabling the user to grab and move items efficiently.

Figure 17 – Codes used in the application for ingredients and tool detection

While Aruco tags have been considered, since in a complete version of the application they would remove the need to print a code for each ingredient and tool, we decided on these custom codes since they allow the user to immediately visualize and understand what he is working with.

Talking about changes not already discussed, originally, we planned to let users purchase the ingredients from a screen named To-Do list, grouping the ingredients according to the exercise they were for. Similarly, the icon with which users were able to add ingredients to this list was a "add to the list" icon, without directly involving a shopping cart. Even though no one formally reported issues with this approach, during the heuristic evaluation we sensed some confusion from the evaluators. For this reason, we've decided to change it to a more streamlined "shopping list" section, with related icons.

Also, we added a Microphone button to the input fields (exercise generation prompt and error description), and decided to initiate the vocal input only while selecting it instead of allowing users to select the text field, since we believed it made clearer to users that the input was done through voice

7.d Screens

Home Screen

Users are greeted with the Home Screen, showing the application logo and the various submenus. From here, users can access the application's settings, their shopping list, and can search for an exercise.

Figure 18 – Home Screen

Exercises can be accessed in three ways: first, with a prompt, that will generate custom exercises based on what the user asked for. Users can repeat the input as many times as they want before sending it, to fix Violation #6 and Violation #9.

Figure 19 – exercise generation flow

Then, in the "Daily Exercises" section, users can access a daily playlist of custom generated exercises, and in the "Completed Exercises" section, which provides access to all previously finished tasks. The section has been renamed to fix Violation #4.

Exercise Selection

Once the "category" of exercises is chosen, users are greeted with a screen showing the possible choices.

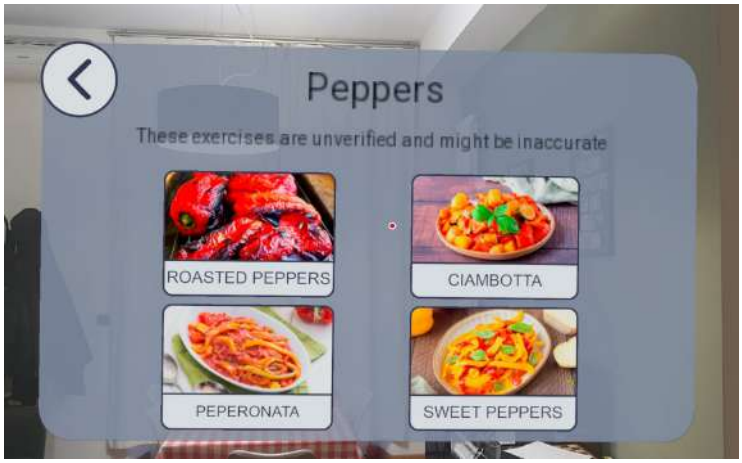


Figure 20 – Exercise Selection Screen

Once an exercise is selected, it's possible to check its details, such as difficulty, time and a description.

Figure 21 – Exercise Details Screen

The description can be expanded, to fix violation #25.

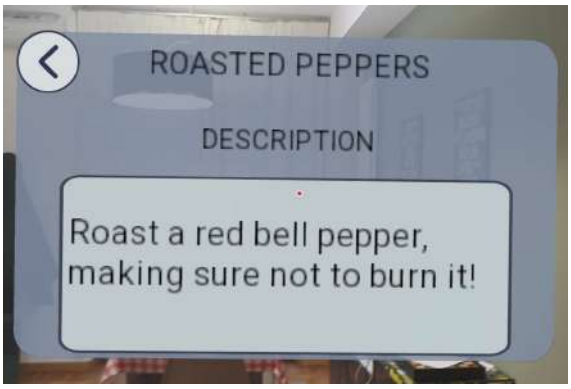


Figure 22 – Expanded exercise details

It's also possible to access the ingredients list, from which it's possible to add them to the "shopping list".

Figure 23– Exercise ingredients

Exercise Execution

Once an exercise is selected, users are by a prompt telling what needs to be done to start the exercise.

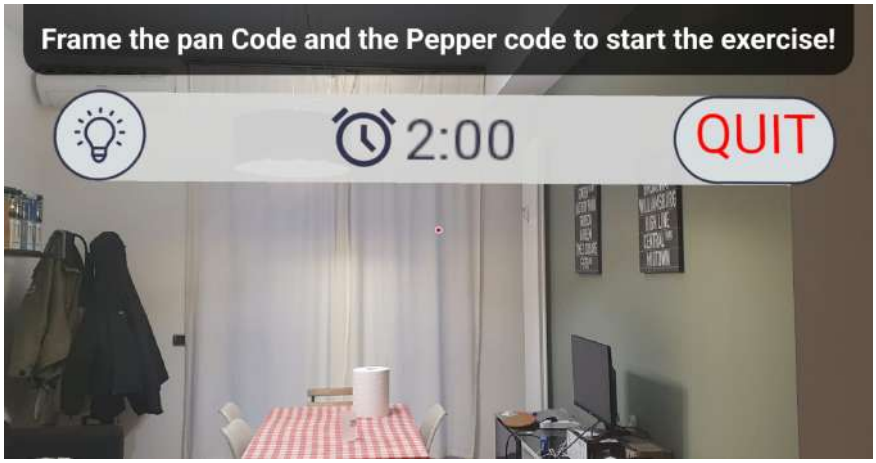


Figure 24 – Exercise start

During the exercise execution, users are provided with a small HUD containing an indicator for the remaining time, a button to quit the exercise and a button to show a hint.

During the exercise, the ingredients and tools are rendered in real time on top of the codes, allowing users to freely experiment and interact with them.

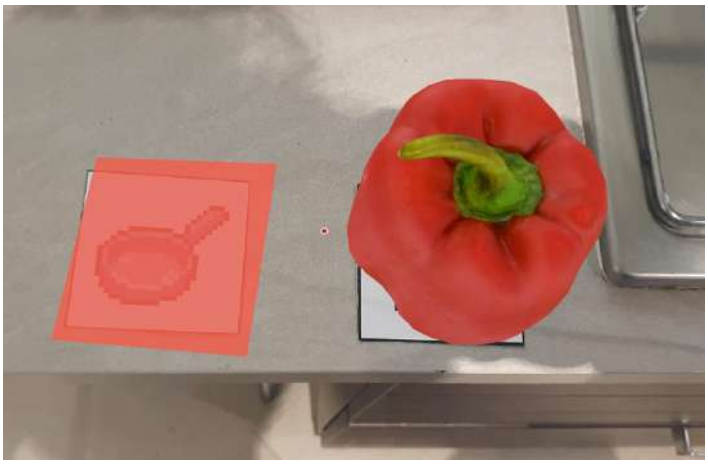


Figure 25 – Rendered pan and pepper

Users have constant visual feedback on what is happening. For example, when an ingredient is cooking the pan starts to smoke

Figure 26 – Pepper smoking once placed on the pan

and when it's done cooking it changes color.

Figure 27 – Cooked pepper

Once completed, the user can see the outcome of the exercise, with it either being a "Success" or a "Fail".

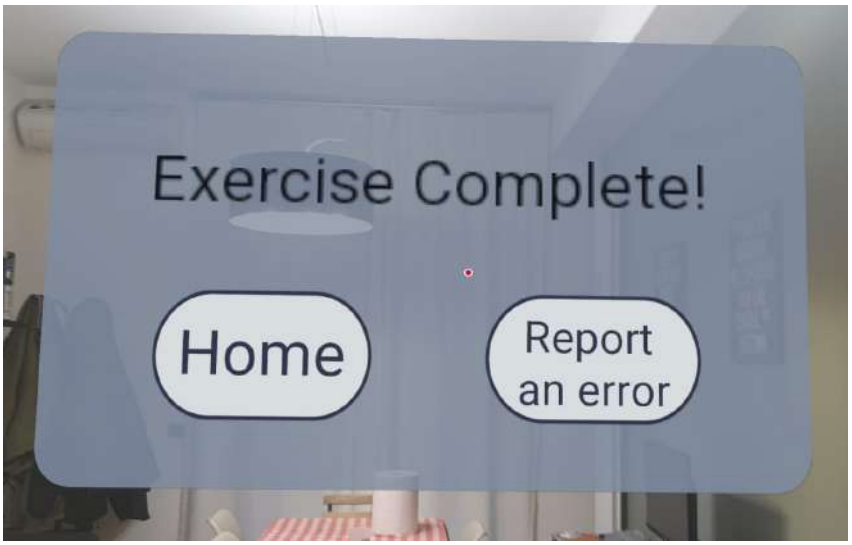


Figure 28 – Exercise Completed Screen

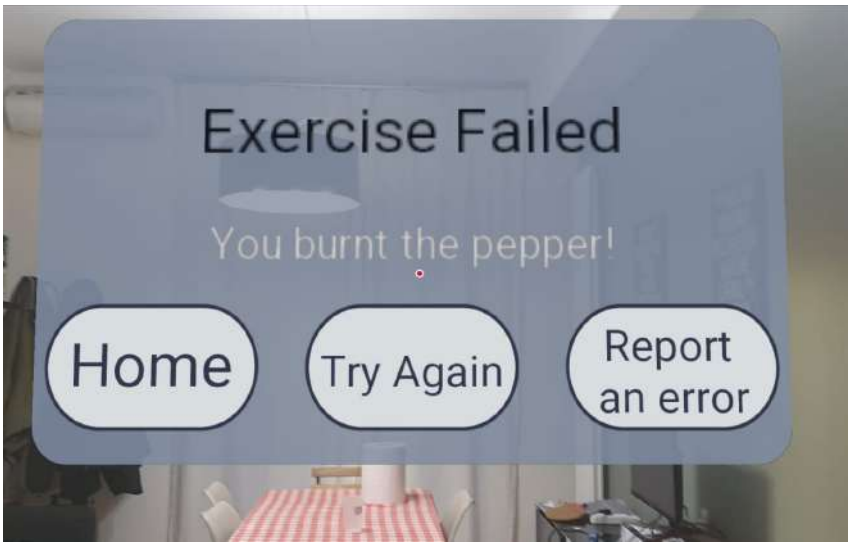


Figure 29 – Exercise Failed Screen

In this second case, the user can also learn the reason for the failure.

Error Reporting

Once an exercise is complete, users can report an error, by using vocal input.

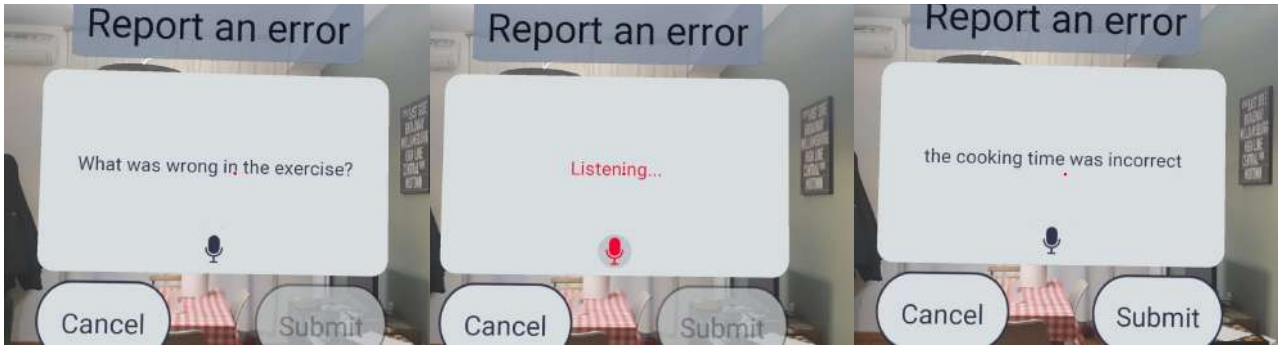


Figure 30 – Error report flow

Once submitted, the application thanks the user and explains the effects of the report, following the Microsoft's guidelines for Human-AI interaction, which state that an AI system must "Immediately update or convey how user actions will impact future behaviors of the AI system".

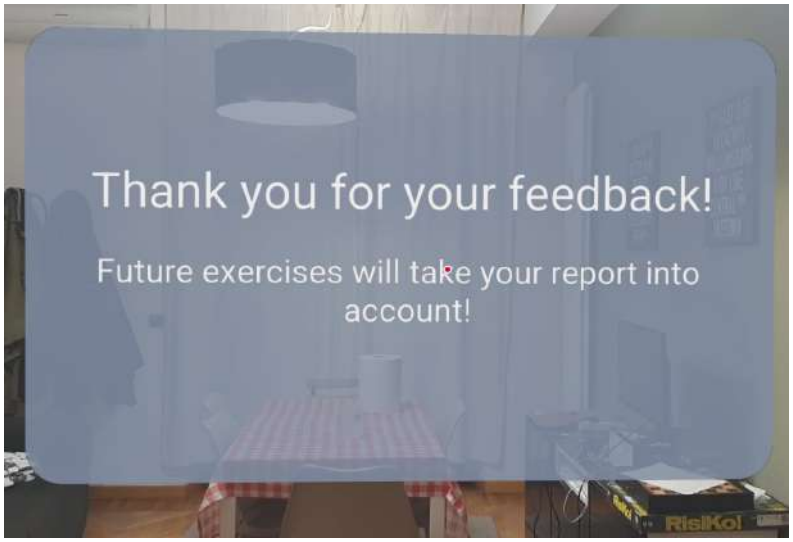


Figure 31 – "Thank you" Screen

Shopping List

From this screen users can add ingredients from various exercises to their Amazon cart.

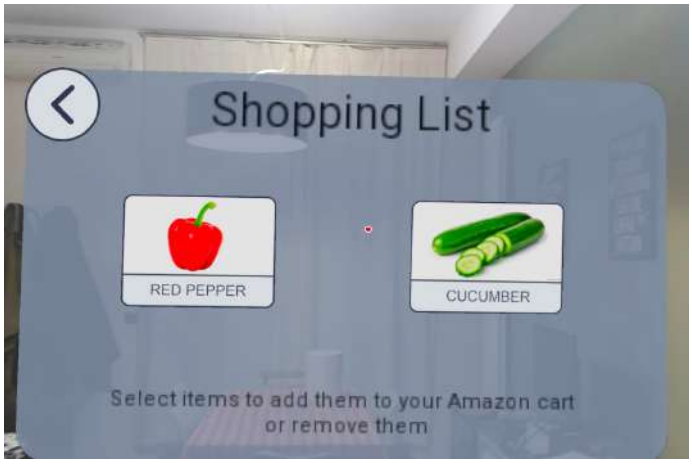


Figure 32 – Shopping List Screen

Users can select individual ingredients and can decide if they want to add the selected items to their cart or remove them from the list.

Figure 33 – Single ingredient selection

The individual selection has been added to fix violation #3.

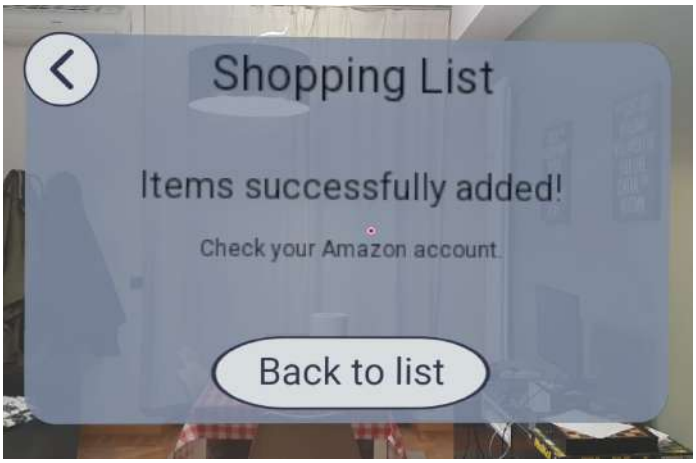


Figure 34 – Feedback for adding item to Amazon cart

The "remove" button has been moved to the top of the screen to fix violations #1 and #19, and its behavior has been changed to fix violation #24.

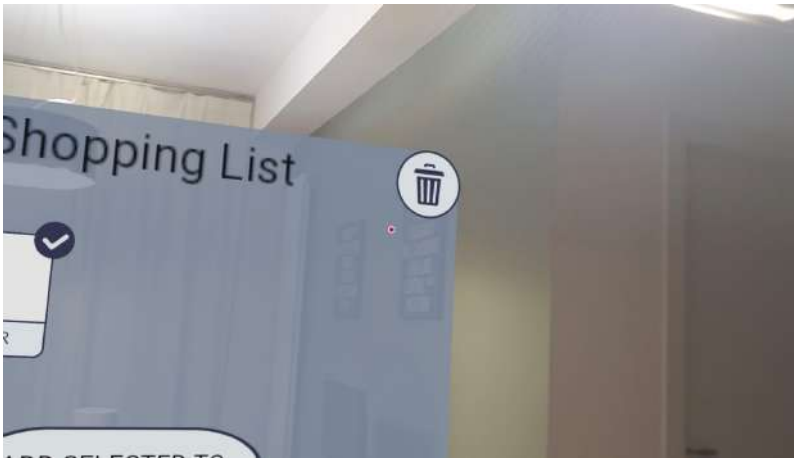


Figure 35 – Remove button

Visual Design

While the framework provided very limited integration with CSS, severely limiting the possibilities of a structured interface design without recurring to 3D models (which no one in the team had experience with), we still made sure to maintain a consistent design between each screen.

We opted for a minimalistic color palette, since we wanted to keep the AR HUD elements as simple and recognizable as possible.

Figure 36 – BlitzCook color palette

For the same reason, we decided to keep a colored background for most of the screens, making sure that users can easily distinguish what is part of the AR application from what is real. We kept the background semi-transparent, allowing users to fully be aware of what is happening in the surrounding environment.

We also designed minimalistic buttons to navigate the application, using icons to ensure consistency.

Figure 37 –BlitzCook buttons

7.e Standards and Constraints of the Target Device

As already explained in the "Design Choices and Changes" section, AR Headsets don't have native input devices to interact with the application, often relying on external peripherals such as keyboards or controllers. Since this solution was not suitable for our application, we used Gaze Input. This approach is widely used in augmented or virtual reality applications. For the same reason we opted for vocal inputs for operations which required text, another solution often used by AR systems.

7.f Limitations

The first and most important limitation concerns the Target Device: while the application has been developed with an AR Headset in mind, it currently is a standard AR mobile application. This solution has been considered as sufficient and suitable for the purpose of the prototype.

For the same limitations of the library, the text is shown at a low resolution most of the time. Thanks to a workaround, it was possible to show high-resolution text inside buttons and in certain sections, but not in the entire app.

The other main limitations of the prototype are related to the use of AI.
The exercise generation is simulated, they are already coded in the prototype and shown according to what the user searches. The "daily exercises" section always shows the same exercises, and the search bar has only two possible outcomes: one where the search contains the word "pepper" and another where it contains the word "chocolate". Moreover, the only accessible exercise is "Roasted Peppers", while the others can be browsed but not started. One exercise has been considered sufficient to show the app's purpose.
The same goes for the "Report an error" function, which does not impact the exercise generation in any way.
However, we made sure to simulate these functionalities following Microsoft's Guidelines for Human-AI Interaction, providing the user with all the information he needs and informing him if his actions have an impact on the future experience.

Returning to the exercises, their execution is partially hard coded: while the idea is that the application can recognize the user's actions and interpret them with the virtual ingredients, the actual implementation would have been too complex. For this reason, the application detects actions by checking if the "pan" code is framed or not, simulating the originally planned interaction for simple exercises concerning cooking times.

For what concerns the Amazon cart, the integration has not been implemented: the prototype handles the ingredients list locally without adding the ingredients to the Amazon cart. The original idea was to implement the Amazon Product Advertising API, but due to limitations with "fresh" products it proved to be impossible.

8. Usability Testing

8.a.i-ii Preparation and Run + links

Participants

Once the High-Fidelity prototype had been completed, we started the usability tests. We reached the following participants thanks to mutual friends:

Participant 1:
Simone, 25, university student
Expertise with AI systems: high, daily usage
Expertise with cooking: moderate knowledge of techniques, tools and ingredients
Paste experience with AR systems: little to none

Participant 2:
Ilaria, 23, university student
Expertise with AI systems: high, daily usage
Expertise with cooking: moderate sufficient knowledge of techniques, tools and ingredients
Paste experience with AR systems: none

Participant 3:
Nicole, 25, elementary school teacher
Expertise with AI systems: high, daily usage
Expertise with cooking: basic knowledge of techniques, tools and ingredient
Paste experience with AR systems: little to none

We made sure to perform the tests with people we didn't know well, to prevent any influence on the evaluations.

Roles Definition

We then decided that Pietro Pantani would be the facilitator, with the other two team members being the observers. The choice of keeping the same roles has been done to ensure consistency between each testing session.

Material

We then proceed with the preparation of the **testing protocol**.
Alongside everything needed to perform the tests and record the results, we decided to use a computer connected to the device to quickly change the status of the app, to make the preparation for the various tasks as fast as possible.

Tasks Definition

We then defined the tasks, breaking the three main ones in 7 smaller subtasks. This was done to define a small story about trying to learn a specific recipe, so that testers could have better context on what they were trying to do.

The tasks are the following:

1. Yesterday you burnt peppers while trying to cook them. You use BlitzCook to do an exercise to practice for that.
2. After completing the exercise, you notice that the cooking time for the peppers was way different from the actual one. You want to report this issue to the application.
3. After a while, you want to retry the exercise about peppers. Try to find the exercise and start it again.
4. While doing an exercise, you are confused and want some help on how to complete it. Try to see a hint.
5. After completing the peppers exercise, you want to try it in real life. You want to add the ingredients needed to practice to your shopping list.
6. Next week, you decide to add the peppers from your shopping list to your Amazon cart.
7. After adding the peppers, you notice that you added other ingredients to your shopping list by mistake. Try to remove the unwanted items.

For what concerns the methodology, we've decided to apply the **think-aloud**, since we wanted to know if the user was correctly understanding the application.

We also defined a list of critical and non-critical errors for each task, which will affect the success score if made.

Evaluation Criteria

After defining the tasks, we decided on which evaluation metrics to adopt.

- Time on task: even though this evaluation metric can be affected by the think-aloud methodology, we still decided to use it to measure if certain sections of the application were "quick" to grasp and understand. We made sure to choose appropriate time intervals for each task, by first measuring the task completion times with other people that were not suitable for being full on testers and then deciding a reasonable time.
- Success rate: for each task, we made a list of possible critical and non-critical errors, each one impacting the final success rate of the task. We started from 100%, and each non-critical error lowered the score by a set value for the error. On the other end, critical errors reduced the success rate to 0%.
- Number of critical errors: for each task, we made a list of possible critical errors, which prevent the task from being considered "completed".
- SEQ score: at the end of each task, we asked participants to rate the difficulty of the task on a scale from 1 (very difficult) to 7 (very easy).

- SUS score: after the completion of the seven tasks, we asked participants to fill a short questionnaire, being Brooke's System Usability Scale.
- Likes, Dislikes and Recommendations: after the testing session, we asked participants some open-ended questions to gain additional feedback, being:
    - What's the thing that you liked the most about our app and why?
    - What's the thing that you liked the least about our app and why?
    - Is there anything you would change or add to make the app easier to use?
    - Did you notice any difficulties with the "pointer" used to navigate the app?
    - Is there anything you would change or add to make the app more useful?

The full testing protocol can be accessed at the following link:
https://github.com/polito-hci-2025/STRFL/blob/main/A5/Testing%20Protocol.pdf

The SUS questionnaire we made the testers fill can be accessed at the following link:
https://github.com/polito-hci-2025/STRFL/blob/main/A5/System%20Usability%20Scale%20(SUS).pdf

We also prepared the consent forms, both in English and in Italian, to ensure that participants understand everything that they are signing without the need of a translation. Since the participants were all Italian, they only signed the Italian version.

The forms can be accessed at the following link:
https://github.com/polito-hci-2025/STRFL/blob/main/A5/InformedConsentForm.pdf

And their signed versions can be accessed at the following link:
https://github.com/polito-hci-2025/STRFL/tree/main/A5/Signed%20Consent%20Form

Execution

Once the planning was done, we set the appointments with the testers.
We were able to perform the tests in the testers' kitchens, allowing us to see the application in action in the most realistic possible settings.

During the evaluation we strictly followed the script, keeping a friendly yet professional tone, making sure to make the testers at ease.

Since the testers were Italian speakers, we conducted the tests in their native language. Additionally, we offered the possibilities of translating the application interface in real time and their vocal inputs, but they were able to use the application without any additional help.

Figures 38, 39 – Usability Tests

While performing the cooking exercise, the facilitator held the smartphone for the testers, keeping their hands free and making sure that they can perform it without any issues, simulating an AR Headset experience as closely as possible.

Figure 40 – Hands free test

8.b.i-ii-iii Results

The tests went very smoothly: the participants were able to complete all of the tasks with little to no error.
We specifically asked participants if they had any experience with AR systems before and, even though they had little to no experience, they were able to navigate the application without any issues.
While hearing their thought process, we learnt that they correctly understood each button and section of the application without needing extra help.

Task Score

TESTER	T1	T2	T3	T4	T5	T6	T7	Average
Ilaria	100%	90%	100%	100%	100%	90%	100%	97,14%
Nicole	100%	100%	100%	100%	100%	90%	100%	98,45%
Simone	100%	100%	100%	100%	100%	100%	60%	94,28%

Each tester completed each task with little to no errors.
The only significant error has been made by Simone, who removed items from the cart by the "Recipe Details" page instead of the "Shopping List" page.

SEQ Score

TESTER	T1	T2	T3	T4	T5	T6	T7	Average
Ilaria	6	5	7	7	7	7	7	6,57
Nicole	6	7	7	6	7	7	5	6,42
Simone	7	7	7	7	6	7	7	6,85

Cumulative Average: 6,61

The post-task questions confirmed our initial impressions: users found little to no difficulties while completing tasks.

SUS Score

TESTER	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Score
Ilaria	4	1	5	2	4	1	5	1	4	1	90
Nicole	4	1	4	1	5	1	4	1	5	1	92,5
Simone	4	1	5	1	5	1	5	1	5	1	97,5

Average Score: 93,33

The post-test questionnaire revealed to us that users liked the application, and most importantly that they didn't need external help to operate it. This point is extremely important, since the combination of AR controls and external printable codes could have been seen as "too technical".

Impressions and potential changes

After the post-test questionnaire, we asked participants for additional feedback.
We learnt that they enjoyed being able to see the virtual ingredients and interact with them and believed that the app could be useful in their daily life.
From these sessions, we learnt that some of the participants would have enjoyed a faster selection with the Gaze Pointer.

While making it faster could be a problem with less experienced users, a potential change could be to add a setting for the pointer speed, allowing users to customize their experience to their liking.

Additionally, some users suggested showing images of the required codes for an exercise in the "ingredients" section, to make it easier to prepare for an exercise before starting it.

Even though no user pointed this out, while further analyzing this suggestion we thought about adding a "tutorial video" or at least a "guide" accessible from the home screen, better explaining how to use the printable codes.

9. Conclusions

9.a Main learnings

During this semester, we really enjoyed learning the whole process described by the course: solving real problems is the main job of engineers, and learning the best ways of interacting with people to understand their needs is something that has not been taught in any other course we attended so far.

Additionally, we really liked the aspects related to application design often overlooked in other courses, like understanding what makes a good visual design and what is understandable for users.

Most importantly, learning how to design systems using AI was interesting, since it's a powerful technology that is getting increasingly popular, and many developers are

integrating it into their application just to follow the trend, without actually understanding what a good use of AI is and how users can interact with it effectively.

The group project was a great opportunity to put all of this into practice, and we were also excited to learn how to design and implement an AR system, something that was new for all of us.

9.b Group Feedback

Regarding group work, all participants actively collaborated from the first assignment, completing all the divided tasks and never failing to attend the shared tasks. Our main strength was dialogue. Every decision was made after approval by the whole group.

To manage our workflow efficiently, we utilized tools like Miro, which helped us keep track of progress and share feedback in real-time. From the beginning, we established a clear division of tasks with set deadlines. Once individual tasks were finished, we always proceeded with a collective review to ensure consistency and quality across the entire assignment.

For the final assignment we had to divide the workload based on needs (exams, personal commitments), giving more freedom to those with upcoming exams.